

CLAIRVOYANCE: A PIPELINE TOOLKIT FOR MEDICAL TIME SERIES

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ABSTRACT

Time-series learning is the bread and butter of data-driven *clinical decision support*, and the recent explosion in ML research has demonstrated great potential in various healthcare settings. At the same time, medical time-series problems in the wild are challenging due to their highly *composite* nature: They entail design choices and interactions among components that preprocess data, impute missing values, select features, issue predictions, estimate uncertainty, and interpret models. Despite exponential growth in electronic patient data, there is a remarkable gap between the potential and realized utilization of ML for clinical research and decision support. In particular, orchestrating a real-world project lifecycle poses challenges in engineering (i.e. hard to build), evaluation (i.e. hard to assess), and efficiency (i.e. hard to optimize). Designed to address these issues simultaneously, Clairvoyance proposes a unified, end-to-end, autoML-friendly pipeline that serves as a (i) software toolkit, (ii) empirical standard, and (iii) interface for optimization. Our ultimate goal lies in facilitating transparent and reproducible experimentation with complex inference workflows, providing integrated pathways for (1) personalized prediction, (2) treatment-effect estimation, and (3) information acquisition. Through illustrative examples on real-world data in outpatient, general wards, and intensive-care settings, we illustrate the applicability of the pipeline paradigm on core tasks in the healthcare journey. To the best of our knowledge, Clairvoyance is the first to demonstrate viability of a comprehensive and automatable pipeline for clinical time-series ML.

Python Software Repository: <https://github.com/vanderschaarlab/clairvoyance>

1 INTRODUCTION

Inference over time series is ubiquitous in medical problems [1–7]. With the increasing availability and accessibility of electronic patient records, machine learning for *clinical decision support* has made great strides in offering actionable predictive models for real-world questions [8, 9]. In particular, a plethora of methods-based research has focused on addressing specific problems along different stages of the clinical data science pipeline, including preprocessing patient data [10, 11], imputing missing measurements [12–16], issuing diagnoses and prognoses of diseases and biomarkers [17–25], estimating the effects of different treatments [26–31], optimizing measurements [32–36], capturing

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uncertainty [37-41], and interpreting learned models [42-46]. On the other hand, these component tasks are often formulated, solved, and implemented as mathematical problems (on their own), resulting in a stylized range of methods that may not acknowledge the complexities and interdependencies within the real-world clinical ML project lifecycle (as a composite). This leads to an often punishing *translational barrier* between state-of-the-art ML techniques and any actual patient benefit that could be realized from their intended application towards clinical research and decision support [47-51].

Three Challenges To bridge this gap, we argue for a more comprehensive, systematic approach to development, validation, and clinical utilization. Specifically, due to the number of moving pieces, managing real-world clinical time-series inference workflows is challenging due the following concerns:

- First and foremost, the *engineering* problem is that building complex inference procedures involves significant investment: Over 95% of work in a typical mature project is consumed by software technicals, and <5% addressing real scientific questions [52]. As a clinician or healthcare practitioner, however, few resources are available for easily developing and validating *complete* workflows. What is desired is a simple, consistent development and validation workflow that encapsulates all major aspects of clinical time-series ML—from initial data preprocessing all the way to the end.
- Second, the *evaluation* problem is that the performance of any component depends on its *context*; for instance, the accuracy of a prediction model is intimately tied to the data imputation method that precedes it [13, 14]. As an ML researcher, however, current empirical practices typically examine the merits of each component individually, with surrounding steps configured as convenient for ensuring “all else equal” conditions for assessing performance. What is desired is a structured, realistic, and reproducible method of comparing techniques that honestly reflects interdependencies in the gestalt.
- Lastly, the *efficiency* problem is that sophisticated designs tend to be resource-intensive to optimize, and state-of-the-art deep learning approaches require many knobs to be tuned. As a clinical or ML practitioner alike, this *computational* difficulty may be compounded by pipeline combinations and the potential presence of temporal distribution shifts in time-series datasets [53]. What is desired is a platform on which the process of pipeline configuration and hyperparameter optimization can be automated—and through which new optimization algorithms to that effect may be built and tested.

Contributions We tackle all three issues simultaneously. The Clairvoyance package is a unified, end-to-end, autoML-friendly pipeline for medical time series. (i) As a *software toolkit*, it enables development through a single unified interface: Modular and composable structures facilitate rapid experimentation and deployment by clinical practitioners, as well as simplifying collaboration and code-sharing. (ii) As an *empirical standard*, it serves as a complete experimental benchmarking environment: Standardized, end-to-end pipelines provide realistic and systematic context for evaluating novelties within individual component designs, ensuring that comparisons are fair, transparent, and reproducible. (iii) Finally, as an *interface for optimization* over the pipeline abstraction, Clairvoyance enables leveraging and developing algorithms for automatic pipeline configuration and stepwise selection, accounting for interdependencies among components, hyperparameters, and time steps. Through illustrative examples on real-world medical datasets, we highlight the applicability of the proposed paradigm within personalized prediction, personalized treatment planning, and personalized monitoring. To the best of our knowledge, Clairvoyance is the first coherent effort to demonstrate viability of a comprehensive, structured, and automatable pipeline for clinical time-series learning.

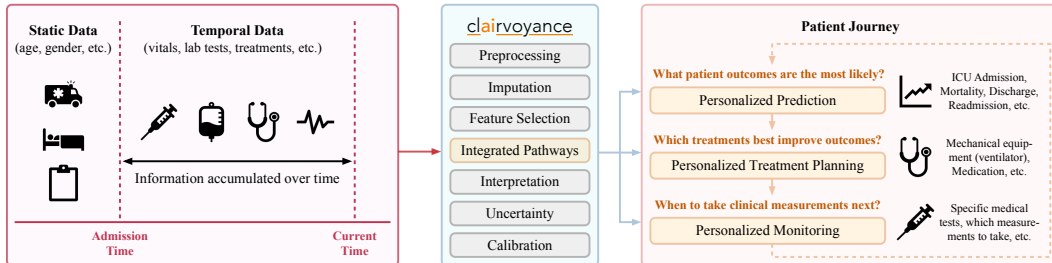


Figure 1: *Clairvoyance and the Patient Journey*. The healthcare lifecycle revolves around asking (1) *what* outcomes are most likely, (2) *which* treatments may best improve them, and (3) *when* taking additional measurements is most informative. Utilizing both static and temporal data, Clairvoyance provides corresponding pathways for personalized prediction of outcomes, personalized estimation of treatment-effects, and personalized monitoring.

2 THE CLAIRVOYANCE PIPELINE

The Patient Journey Consider the typical patient’s interactions with the healthcare system. Their healthcare lifecycle revolves tightly around (1) forecasting outcomes of interest (i.e. the prediction problem), (2) selecting appropriate interventions (i.e. the treatment effects problem), and (3) arranging followup monitoring (i.e. the active sensing problem). Each of these undertakings involve the full complexity of preparing, modeling, optimizing, and drawing conclusions from clinical time series. Clairvoyance provides model pathways for these core tasks in the patient journey (see Figure 1)—integrated into a single pipeline from start to finish (see Figure 2). Formally, these pathways include:

- **Predictions Path.** Let $\{(s_n, \mathbf{x}_{1:T_n})\}_{n=1}^N$ denote any medical time-series dataset, where s_n is the vector of static features for the n -th patient, and $\mathbf{x}_{1:T_n} \doteq \{\mathbf{x}_{n,t}\}_{t=1}^{T_n}$ is the vector sequence of temporal features. *One-shot* problems seek to predict a vector of labels $\mathbf{y}_n$ from $(s_n, \mathbf{x}_{n,1:T_n})$: e.g. prediction of mortality or discharge, where $y_n \in \{0, 1\}$. *Online* problems predict some target vector $\mathbf{y}_{n,t}$ from $(s_n, \mathbf{x}_{n,1:t})$ at every time step: e.g. τ -step-ahead prediction of biomarkers $\mathbf{y}_{n,t} \subseteq \mathbf{x}_{n,t+\tau}$.
- **Treatment Effects Path.** For individualized treatment-effect estimation [26–31], we additionally identify interventional actions $\mathbf{a}_{n,t} \subseteq \mathbf{x}_{n,t}$ at each time step (e.g. the choices and dosages of prescribed medication), as well as corresponding measurable outcomes $\mathbf{y}_{n,t} \subseteq \mathbf{x}_{n,t+\tau}$. The learning problem now consists in quantifying the (factual or counterfactual) potential outcomes $\mathbf{y}_{n,t+\tau}$ that would result from any specific sequence of interventions and patient covariates $(s_n, \mathbf{x}_{n,1:t}, \mathbf{a}_{n,1:t})$.
- **Active Sensing Path.** In addition to mapping (already-measured) covariates to targets, the very decision of what (and when) to measure is also important under resource constraints. In medical settings, active sensing deals with balancing this trade-off between information gain and acquisition costs [32–36]. With reference to some downstream task (e.g. predicting $\mathbf{y}_{n,t+1}$), the aim is to select a subset of covariates $\mathcal{K}_{n,t}$ at each t to maximize the (net) benefit of observing $\{x_{n,t,k}\}_{k \in \mathcal{K}_{n,t}}$.

As a Software Toolkit Engineering *complete* medical time-series workflows is hard. The primary barrier to collaborative research between ML and medicine seldom lies in any particular algorithm. Instead, the difficulty is operational [6, 48, 54]—i.e. in coordinating the entire data science process, from handling missing/irregularly sampled patient data all the way to validation on different popu-

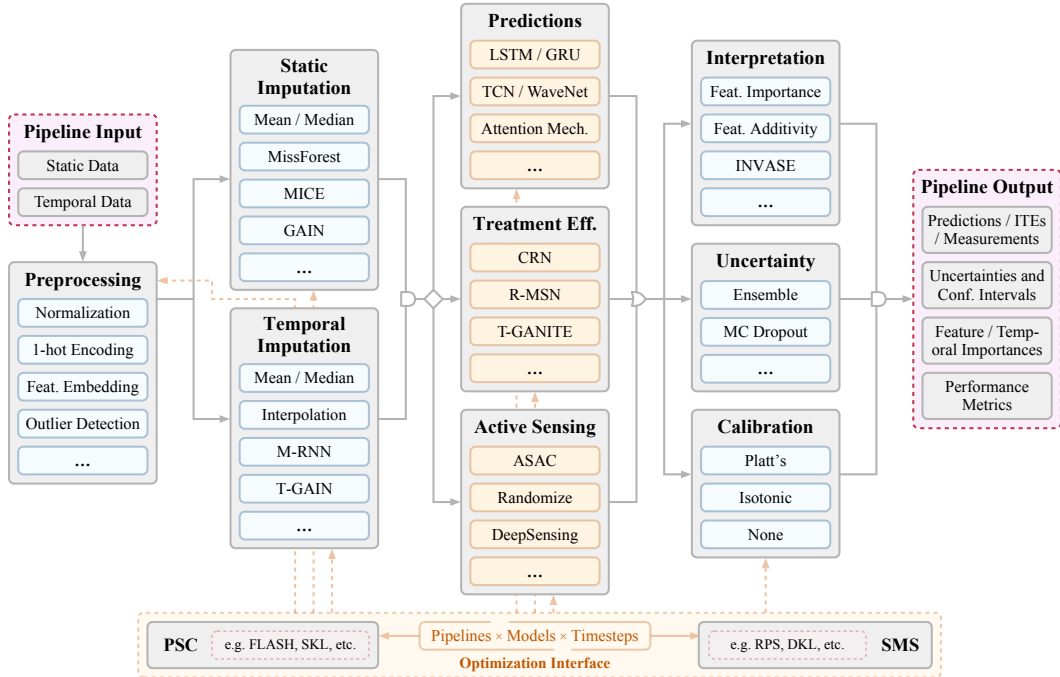


Figure 2: *Clairvoyance Pipeline Overview.* (Dashed) purple cells denote pipeline inputs/outputs, and (solid) gray cells denote pipeline components. Orange options give main pathway models, and blue options give surrounding components. (Solid) gray arrows indicates pipeline workflow, and (dashed) orange the optimization interface.

```

"""Configure Data Preprocessing"""
preprocessing = PipelineComposer(
    FilterNegative(...), OneHotEncoder(...),
    Normalizer(...), ...)

"""Configure Problem Specification"""
specification = ProblemMaker(
    problem_class='online', max_seq_len=24,
    label=['ventilator'], treatment=None, window=4, ...)

"""Configure Data Imputation"""
imputation = PipelineComposer(
    Imputation(type='static', model_name='...', ...),
    Imputation(type='temporal', model_name='...', ...))

"""Configure Feature Selection"""
feature_selection = PipelineComposer(
    FeatureSelection(type='static', model_name='...', ...),
    FeatureSelection(type='temporal', model_name='...', ...))

"""Configure Pathway Model"""
prediction_model = Prediction(model_name='...',
    parameter_dict={}, ...)

"""Load Datasets"""
data_train, data_test = DataLoader.load(
    static_dir='...', temporal_dir='...', ...),
    DataLoader.load(static_dir='...', temporal_dir='...')

"""Execute Pipeline"""
for component in [preprocessing,
    specification,
    imputation,
    feature_selection]:
    data_train = component.fit_transform(data_train)
    data_test = component.transform(data_test)

prediction_model.fit(data_train, ...)
test_output = prediction_model.predict(data_test, ...)

```

Figure 3: *Illustrative Usage*. A prototypical structure of API calls for constructing a prediction pathway model. Clairvoyance is modularized to abide by established fit/transform/predict design patterns. (Green) ellipses denote additional configuration; further modules (treatments, sensing, uncertainty, etc.) expose similar interfaces.

lations [4, 55-60]. Clairvoyance gives a single *unified* roof under which clinicians and researchers alike can readily address such common issues—with the only requirement that the data conform to the standard EAV open schema for clinical records (i.e. patient key, timestamp, parameter, and value).

Under a simple, consistent API, Clairvoyance encapsulates all major steps of time-series modeling, including (a.) loading and (b.) preprocessing patient records, (c.) defining the learning problem, handling missing or irregular samples in both (d.) static and (e.) temporal contexts, (f.) conducting feature selection, (g.) fitting prediction models, performing (h.) calibration and (i.) uncertainty estimation of model outputs, (j.) applying global or instance-wise methods for interpreting learned models, (k.) computing evaluation metrics, and (l.) visualizing results. Figure 2 shows a high-level overview of major components in the pipeline, and Figure 3 shows an illustrative example of usage.

All component modules are designed around the established *fit-transform-predict* paradigms, and the modeling workflow is based around a single chain of API calls. In this manner, each stage in the pipeline is *extensible* with little effort: Novel techniques developed for specific purposes (e.g. a new state-of-the-art imputation method) can be seamlessly integrated via simple wrappers (see Appendix G for an example of how this can be done for any existing method, e.g. from sklearn). This stepwise composability aims to facilitate rapid experimentation and deployment for research, as well as simplifying collaboration and code-sharing. Package documentation/tutorials give further software details.

As an Empirical Standard Evaluating any algorithm depends on its *context*. For instance, how well a proposed classifier ultimately performs is invariably coupled with the upstream feature-selection method it is paired with [44]. Likewise, the accuracy of a state-of-the-art imputation method cannot be assessed on its own: With respect to different downstream prediction models, more sophisticated imputation may actually yield inferior performance relative to simpler techniques [13, 14]—especially if components are not jointly optimized [15]. While current research practices typically seek to isolate individual gains through “all-else-equal” configurations in benchmarking experiments, the degree of actual overlap in pipeline configurations *across* studies is lacking: There is often little commonality in the datasets used, preprocessing done, problem types, model classes, and prediction endpoints. This dearth of *empirical standardization* may not optimally promote practical assessment/reproducibility, and may obscure/entangle true progress. (Tables 6-7 in Appendix A give a more detailed illustration).

Clairvoyance aims to serve as a *structured* evaluation framework to provide such an empirical standard. After all, in order to be relevant from a real-world medical standpoint, assessment of any single proposed component (e.g. a novel ICU mortality predictor) can—and should—be contextualized in the entire *end-to-end* workflow as a whole. Together, the ‘problem-maker’, ‘pipeline-composer’, and all the pipeline component modules aim to simplify the process of specifying, benchmarking, and (self-)documenting full-fledged experimental setups for each use case. At the end of the day, while results from external validation of is often heterogeneous [2, 59, 61], improving transparency and reproducibility greatly facilitates code re-use and independent verification [54, 56, 57]. Just as the “environment” abstraction in OpenAI Gym does for reinforcement learning, the “pipeline” abstraction in Clairvoyance seeks to promote accessibility and fair comparison as pertains medical time-series.

(a) *Example: SASH ‘decomposed’ as SMS (fulfilled here by DKL) followed by combiner (stacking ensemble):*

```
stepwise_pathway_models = []

"Optimize Each Class"
for klass in list_of_pathway_classes:
    sms_agent = Stepwise(method='dkl',
                        klass, data_train, metric)
    models, scores = sms_agent.optimize(num_iters=300)

    sms_model = StepwiseEnsemble(models, scores)
    stepwise_pathway_models.append(sms_model)

"Ensemble Over Classes"
for model in stepwise_pathway_models:
    for step in stepwise_model:
        step.load_model(step.get_path(step.model_id))

pathway_model = StackingEnsemble(stepwise_pathway_models)
pathway_model.fit(data_train, ...)
test_output = pathway_model.predict(data_test, ...)
```

(b) *Example: SPSC ‘decomposed’ as PSC (fulfilled here by SKL) followed by SMS (fulfilled here by DKL):*

```
pipeline_classes = [list_of_static_imputation_classes,
                    ..., ..., ..., list_of_pathway_classes]

"PSC Optimization"
psc_agent = Componentwise(method='skl',
                        pipeline_classes, data_train, data_test, metric)
components, score = psc_agent.optimize(num_iters=300)

pathway_class, data_train, data_test = \
    psc_agent.get_pathway_class_and_data()

"SMS Optimization"
sms_agent = Stepwise(method='dkl',
                    pathway_class, data_train, metric)
models, scores = sms_agent.optimize(num_iters=300)

pathway_model = StepwiseEnsemble(models, scores)
pathway_model.load_model(...)
test_output = pathway_model.predict(data_test, ...)
```

Figure 4: *Optimization Interface*. Example code using the optimization interface to conduct stepwise (i.e. across time steps) and componentwise (i.e. across the pipeline) configuration. Each interface is implementable by any choice of new/existing algorithms. The DKL implementation of SMS is provided for use the Section 4 examples.

As an Optimization Interface Especially in cross-disciplinary clinical research—and during initial stages of experimentation—automated optimization may alleviate potential scarcity of expertise in the specifics of design and tuning. The Clairvoyance pipeline abstraction serves as a software *interface* for optimization algorithms—through which new/existing techniques can be applied, developed, and tested in a more systematic, realistic setting. In particular, by focusing on the temporal aspect of medical time series, this adds a new dimension to classes of autoML problems.

Briefly (see Figure 5), consider the standard task of hyperparameter optimization (for a given model) [62]. By optimizing over classes of *algorithms*, the combined algorithm selection and hyperparameter optimization (“CASH”) problem [63–65] has been approached in healthcare settings by methods such as progressive sampling, filtering, and fine-tuning [50, 66]. By further optimizing over combinations of *pipeline* components, the pipeline selection and configuration (“PSC”) problem [67] has also been tackled in clinical modeling via such techniques as fast linear search (“FLASH”) [68] and structured kernel learning (“SKL”) [67, 69]. Now, what bears further emphasis is that for clinical time series, the *temporal* dimension is critical due to the potential for temporal distribution shifts within time-series data—a common phenomenon in the medical setting (we refer to [53, 70, 71] for additional background). Precisely to account for such temporal settings, the stepwise model selection (“SMS”) problem [71] has recently been approached by such methods as relaxed parameter sharing (“RPS”) [53] as well as deep kernel learning (“DKL”) [71, 72]. Further, what the pipeline interface also does is to naturally allow extending this to define the stepwise algorithm selection and hyperparameter optimization (“SASH”) problem, or even—in the most general case—the stepwise pipeline selection and configuration (“SPSC”) problem. Although these latter two are new—and clearly hard—problems (with no existing solutions), Figure 4 shows simple examples of how the interface allows minimally adapting the SMS and PSC sub-problems (which do have existing solutions) to form feasible (approximate) solutions.

Two distinctions are due: First, Clairvoyance is a pipeline toolkit, not an autoML toolkit. It is not our goal to (re-)implement new/existing optimization algorithms—which abound in literature. Rather, the standardized *interface* is precisely what enables existing implementations to be plugged in, as well as allowing new autoML techniques to be developed and validated within a realistic medical pipeline. All that is required, is for the optimizing agent to expose an appropriate ‘optimize’ method given candidate components, and for such candidates to expose a ‘get-hyperparameter-space’ method. Second—but no less importantly—we must emphasize that we are not advocating *removing* human oversight from the healthcare loop. Rather, the pipeline simply encourages *systematizing* the initial development stages in clinical ML, which stands to benefit from existing literature on efficient autoML techniques.

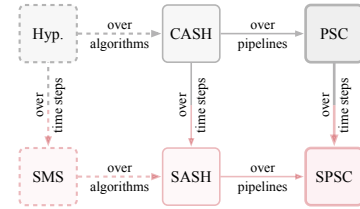


Figure 5: *Degrees of Optimizations*. Clairvoyance allows optimizing over algorithms, pipelines, and time steps.

Design Principles

Our philosophy is based on the authors’ experience in prototyping and developing real-world collaborative projects in clinical time-series. • **Pipeline First, Models Second:** Our first emphasis is on *reproducibility*: The process of engineering and evaluating complete medical time-series workflows needs to be clear and transparent. Concretely, this manifests in the strict “separation of concerns” enforced by the high-level API of each component module along the pipeline (see e.g. Figure 3). With the ‘problem-maker’ and ‘problem-composer’ as first-class objects, the central abstraction here is the pipeline itself, while the intricacies and configurations of individual model choices (e.g. a specific deep learning temporal imputation method) are limited to within each component module. • **Be Minimal and Unintrusive:** Our second emphasis is on *standardization*: While workflow development needs to be unified and systematic, learning to use the framework should be intuitive as well. Concretely, this manifests in the API’s adherence to the existing and popular ‘fit-transform-predict’ paradigm (see e.g. sklearn) in all component modules—both ‘along’ the pipeline steps, as well as ‘across’ the pathways that define the patient’s healthcare lifecycle (see Figure 2). This enables easy adoption and rapid prototyping—qualities that are paramount given the degree of collaborative research and cross-disciplinary code-sharing required in healthcare-related research. • **Encourage Extension:** Our third emphasis is on *extensibility*: Given that novel methods are proposed in the ML community every day, the pipeline components should be easily extensible to incorporate new algorithms. Concretely, this manifests in the encapsulated design for models within each component module: Specifically, in order to integrate a new component method (e.g. from another researcher’s code, or from an external package) into the framework, all that is required is a simple wrapper class that implements the ‘fit’, ‘predict’, and ‘get-hyperparameter-space’ methods; likewise, for an optimization agent (see subsection on optimization interface below), all that is required is to expose an ‘optimize’ method.

Worked Examples

For a discussion of the choice of *built-in* techniques to include with the initial release, see Appendix C. Appendix E gives a worked example of using the Clairvoyance **pipeline** to train and use a model in a standard setting (for this, we use the predictions pathway). Appendix F gives a worked example of using the **optimization** interface to perform stepwise model selection (for this, we use the treatment effects pathway for variety). Appendix G gives an example of how a generic **wrapper** can be written for integrating an external model/algorithm that is not already implemented in the current version of Clairvoyance. Finally, the software repository contains Jupyter notebooks and top-level API code with examples of pathways and optimization.

3 RELATED WORK

Clairvoyance is a pipeline toolkit for medical time-series machine learning research and clinical decision support. As such, this broad undertaking lies at the intersection of three concurrent domains of work: Time-series software development, healthcare journey modeling, and automated learning.

Time-series Software First and foremost, Clairvoyance is a software toolkit. Focusing on challenges common to clinical time-series modeling, it is primarily differentiated by the *breadth* and flexibility of the pipeline. While there exists a variety of sophisticated time-series packages for different purposes, they typically concentrate on implementing collections of algorithms and estimators for specific types of problems, such as classification [79], forecasting [77], feature extraction [76], reductions between tasks [78], or integrating segmentation and transforms with estimators [75]. By contrast, our focus is orthogonal: Clairvoyance aims at end-to-end development along the entire inference workflow, including pathways and pipeline components important to medical problems (see Table 1). Again indeed—if so desired, and as mentioned above—specific algorithms from [73–79] can be integrated into Clairvoyance workflows through the usual ‘fit-transform-predict’ interface, with little hassle.

Healthcare Lifecycle For specific use cases, clearly a plethora of research exists in support of issuing diagnoses [22–24], prognostic modeling [17–21], treatment-effect estimation [26–31], optimizing measurements [32–36], among much more. The key proposition that Clairvoyance advances is the underlying *commonality* across these seemingly disparate problems: It abstracts and integrates along the time-series inference workflow, across outpatient, general wards, and intensive-care environments, and—above all—amongst a patient’s journey of interactions through the healthcare system that call for decision support in predictions, treatments, and monitoring (Figure 1). Now, it also important to state what Clairvoyance is *not*: It is not an exhaustive list of algorithms; the pipeline includes a collection of popular components, and provides a standardized interface for extension. It is also not a solution to preference-/application-specific considerations: While issues such as data cleaning, algorithmic fairness, and privacy and heterogeneity are important, they are beyond the scope of our software.

Table 1: *Clairvoyance and Comparable Software*. *Note that vernacularly, “pipelining” simply refers to the *procedural* workflow (i.e. from inputs to training, cross-validation, outputs, and evaluation); existing packages focus on implementing algorithms for prediction models alone, with minimal preprocessing. In contrast, Clairvoyance provides support along the *data science* pipeline, and across different *healthcare pathways* requiring decision support.

	cesium [73]	tslearn [74]	seglearn [75]	tsfresh [76]	pysf [77]	sktime [78]	pyts [79]	Clairvoyance
Preprocessing	✓	✓	✓	✓	✗	✓	✓	✓
Temporal Imputation	✓	✓	✓	✓	✓	✗	✓	✓
Feature Selection	✗	✗	✗	✓	✗	✗	✗	✓
Predictions	Static Features	✗	✓	✗	✗	✗	✗	✓
	Online Targets	✗	✓	✗	✓	✗	✓	✓
	Predictions	✓	✓	✓	✓	✓	✓	✓
	Treatment Effects	✗	✗	✗	✗	✗	✗	✓
	Active Sensing	✗	✗	✗	✗	✗	✗	✓
Interp.	Feat. Importance	✗	✗	✗	✗	✗	✗	✓
	Feat. Additivity	✗	✗	✗	✗	✗	✗	✓
	Instance-wise	✗	✗	✗	✗	✗	✗	✓
Uncertainty	✗	✗	✗	✗	✗	✗	✗	✓
Calibration	✗	✗	✗	✗	✗	✗	✗	✓
End-to-End Pipelining*	✗	✗	✗	✗	✗	✗	✗	✓
Optimization Interface	✗	✗	✗	✗	✗	✗	✗	✓

Automated Learning Finally, tangentially related is the rich body of work on autoML for hyperparameter optimization [62], algorithm/pipeline configuration [63–65, 67], and stepwise selection [71], as well as specific work for healthcare data [50, 53, 66–68, 70, 71]. In complement to these threads of research, the Clairvoyance pipeline interface enables—if so desired—leveraging existing implementations, or validating novel ones—esp. in efficiently accounting for the temporal dimension.

4 ILLUSTRATIVE EXAMPLES

Recall the patient’s journey of interactions within the healthcare system (Figure 1). In this section, our goal is to illustrate *key usage scenarios* for Clairvoyance in this journey—for personalized (1) prediction, (2) treatment, and (3) monitoring—in outpatient, general wards, and intensive-care environments.

Specifically, implicit in all examples is our proposition that: (i) as a software toolkit, constructing an end-to-end solution to each problem is *easy*, *systematic*, and *self-documenting*; (ii) as an empirical standard, evaluating collections of models by varying a single component ensures that comparisons are *standardized*, *explicit*, and *reproducible*; and (iii) as an optimization interface, the flexibility of selecting over the *temporal dimension*—in and of itself—abstracts out an interesting research avenue.

Medical Environments Our choices of time-series environments are made to reflect the heterogeneity of realistic use cases envisioned for Clairvoyance. For the outpatient setting, we consider a cohort of patients enrolled in the UK Cystic Fibrosis Registry (**CYSTIC**) [80], which records longitudinal follow-up data for ~5,800 individuals with the disease. On the registry, individuals

Medical Environment	Outpatient	General Wards	Intensive Care
Dataset	UKCF [80]	WARDS [81]	MIMIC [82]
Duration of Trajectories	Avg. ~5.3 years	Avg. ~9.1 days	Avg. ~85.4 hours
Variance (25%–50%–75%)	(4–6–7 years)	(6–9–15 days)	(27–47–91 hours)
Frequency of Measurements	Per 6 months	Per 4 hours	Per 1 hour
Different Types of Static and Temporal Features	Demo., Comorbidities, Infections, Treatments	Admiss. Stats, Vital Signs, Lab Tests	Demo., Vital Signs, Lab Tests, Medications
Dimensionality of Features	11 static, 79 temporal	8 static, 37 temporal	11 static, 40 temporal
Number of Samples	~5,800 patients	~6,300 patients	~23,100 patients
Endpoints (cf. Predictions)	FEV1 Result	Admission to ICU	Mechanical Ventilation
Class-label Imbalance	(Continuous-valued)	~5.0%-to-95.0%	~36.8%-to-63.2%

Table 2: *Medical Environments*. We consider the range of settings, incl. outpatient, general wards, and ICU data.

are *chronic patients* monitored over infrequent visits, and for which long-term decline is generally expected. For the general wards setting, we consider a cohort of $\sim 6,300$ patients hospitalized in the general medicine floor in the Ronald Reagan Medical Center (**WARDS**) [81]. In contrast, here the population of patients presents with a wide variety of conditions and diagnoses (1,600+ ICD-9 codes), and patients are monitored more frequently. The data is highly non-stationary: on the hospital floor, deterioration is an *unexpected* event. For the intensive-care setting, we consider $\sim 23,100$ individuals from the Medical Information Mart for Intensive Care (**MIMIC**) [82]. Here, the setting is virtually that more or less “anything-can-happen”, and physiological data streams for each patient are recorded extremely frequently. Varying across the set of environments are such characteristics as the average durations of patient trajectories, the types of static and longitudinal features recorded, their frequencies of measurement, and their patterns and rates of missingness (Table 2 presents some brief statistics).

Example 1 (Lung Function in Cystic Fibrosis Patients) The most common genetic disease in Caucasian populations is cystic fibrosis [83], which entails various forms of dysfunction in respiratory and gastrointestinal systems, chiefly resulting in progressive lung damage and recurrent respiratory infections requiring antibiotics—and in severe cases may require hospitalization and even mechanical ventilation in an ICU (see Example 2) [84, 85]. While classical risk scores and survival models utilize only a fraction of up-to-date measurements, recent work has leveraged deep learning to incorporate greater extents of longitudinal biomarkers, comorbidities, and other risk factors [86]. An essential barometer for anticipating the occurrence of respiratory failures is the gauge of lung function by *forced expiratory volume* (FEV1): Accurate prediction yields an important tool for assessing severity of a patient’s disease, describing its onset/progression, and as an input to treatment decisions [85, 87].

This is an archetypical rolling-window time-series problem for Clairvoyance’s *predictions pathway*. Consider the models in Table 3: (i) As a clinical professional, it goes without saying that building the pipeline for each—or extending additional models through wrappers—has a low barrier to entry (see Figure 3/tutorials/documentation). (ii) As an ML researcher, one can rest assured that such comparisons are expressly standardized: Here, all results are explicitly from same pipeline using min-max normalized features, GAIN for static missing values, M-RNN for temporal imputation, no feature selection, and each model class shown. (iii) Lastly, to highlight the utility of the interface for selection over time, the final row presents results of approaching SASH using the example method of Figure 4(a), and—for fair comparison—with the pipeline kept constant. This simple approach already yields some gains in performance, laying a precedent—and the pipeline infrastructure—for further research.

<i>Dataset (Label)</i>	UKCF (FEV1 Result)		WARDS (Admission to ICU)		MIMIC (Mech. Ventilation)	
Evaluation	RMSE	MAE	AUC	APR	AUC	APR
Attention	(N/A)	(N/A)	0.888 ± 0.016	0.551 ± 0.024	(N/A)	(N/A)
RNN-GRU	0.064 ± 0.001	0.035 ± 0.001	0.865 ± 0.010	0.487 ± 0.048	0.898 ± 0.001	0.774 ± 0.002
RNN-LSTM	0.062 ± 0.001	0.033 ± 0.001	0.841 ± 0.014	0.412 ± 0.032	0.901 ± 0.001	0.776 ± 0.002
Temporal CNN	0.120 ± 0.004	0.096 ± 0.003	0.826 ± 0.020	0.319 ± 0.048	0.884 ± 0.004	0.749 ± 0.007
Transformer	0.081 ± 0.002	0.050 ± 0.002	0.846 ± 0.006	0.472 ± 0.045	0.889 ± 0.002	0.761 ± 0.004
Vanilla RNN	0.070 ± 0.001	0.043 ± 0.001	0.794 ± 0.018	0.277 ± 0.063	0.898 ± 0.001	0.771 ± 0.002
SASH	0.059 ± 0.001	0.030 ± 0.001	0.891 ± 0.011	0.557 ± 0.031	0.917 ± 0.006	0.809 ± 0.013

Table 3: *Predictions Pathway Example*. In addition to (online) 6-month ahead predictions of FEV1 in UKCF, we also test (one-shot) predictions of admission to ICU after 48 hours on the floor in WARDS, and (online) 4-hours ahead predictions of the need for mechanical ventilation in MIMIC (these are extended to treatment and sensing problems below). As the WARDS prediction is one-shot, what is denoted ‘SASH’ for that excludes the SMS ensembling step. Note that the canonical attention mechanism does not permit (variable-length) online predictions.

Example 2 (Mechanical Ventilation on Intensive Care) Mechanical ventilation is an invasive, painful, and extremely unpleasant therapy that requires induction of artificial coma, and carries a high risk of mortality [88]. It is also expensive, with a typical ICU ventilator admission $> \$30,000$ [89]. To the patient, the need for mechanical ventilation—due to evidence of respiratory/ventilatory failure—is by itself an adverse outcome, and is unacceptable to some, even if it means they will not survive. It is possible that alternative strategies employed earlier may alleviate the need for ventilation, such as high flow oxygen, non-invasive ventilation, or—in this example—appropriate *use of antibiotics* [88]. Now, little is known about optimal timing of courses of antibiotics; in most cases a routine number of days is simply chosen when blood is typically sterile after first dose. On the one hand, there is a clear biologically plausible mechanism for incompletely treated infection to lead to longer periods of

critical care, esp. requiring ventilation. On the other hand, antibiotic stewardship is crucial: Over-use of broad spectrum antibiotics leads to resistance, and is by itself a global health emergency [90].

This is an archetypical problem for the *treatment effects pathway*. Table 4 shows the performance of the two state-of-the-art models for estimating effects of treatment decisions over time while adjusting for time-dependent confounding—that is, since actions taken in the data may depend on time-varying variables related to the outcome of interest [30, 31]. We refrain from belaboring points (i), (ii), (iii) above but their merits should be clear. From the *patient’s* perspective, accurate estimation of the effect of treatment decisions on the risk of ventilation may assist them and their carers in achieving optimal shared decision-making about the care that they would like to receive. From the *hospital’s* perspective, many ICUs around the world operate at $\sim 100\%$ bed occupancy, and delayed admission is typically an independent predictor of mortality [91–94]; therefore accurate estimation of the need for escalation or continued ICU ventilation is logistically important for resource planning and minimization of delays.

<i>Time Horizon</i>	Estimating 1 Day Ahead		Estimating 2 Days Ahead		Estimating 3 Days Ahead	
Evaluation	AUC	APR	AUC	APR	AUC	APR
RMSN	0.860 ± 0.005	0.889 ± 0.007	0.790 ± 0.004	0.883 ± 0.003	0.726 ± 0.015	0.852 ± 0.009
CRN	0.865 ± 0.003	0.892 ± 0.004	0.783 ± 0.009	0.872 ± 0.013	0.767 ± 0.010	0.869 ± 0.007
SASH	0.871 ± 0.007	0.902 ± 0.005	0.792 ± 0.003	0.885 ± 0.009	0.771 ± 0.005	0.873 ± 0.003

Table 4: *Treatment Effects Pathway Example*. Results for estimation over different horizon lengths. Note that this uses a $\sim 6,000$ -patient subset (from those in Table 2) who received antibiotics at any point, based on daily decisions on antibiotic treatment, over spans of up to 20 days, with labels distributed 58.9%-to-41.1% overall.

Example 3 (Clinical Deterioration of Ward Patients) Given the delay-critical nature of ICU admission w.r.t. morbidity/mortality, what is often desired is an automated prognostic decision support system to monitor ward patients and raise (early) alarms for impending admission to ICU (as a result of clinical deterioration) [25, 94, 95]. However, observations are costly, and the question of what (and when) to measure is by itself an *active choice* under resource constraints [32–36]: For instance, there is less reason to measure a feature whose value can already be confidently estimated on the basis of known quantities, or if its value is not expected to contribute greatly to the prognostic task at hand.

This is an archetypical problem for Clairvoyance’s *active sensing pathway*. Table 5 indicates the performance of different models for balancing this trade-off between information gain and acquisition rate with respect to admissions to ICU of ward patients. At various budget constraints (i.e. amounts of measurements permitted), each active sensing model learns from the training data to identify the most informative features to measure at test-time, so as to maximize the performance of admission predictions. (To allow some measurements to be costlier than others, they can simply be up-weighted when computing the budget constraint). As before, our propositions (i), (ii), and (iii) are implicit here.

<i>Measure Rate</i>	With 50% Measurements		With 70% Measurements		With 90% Measurements	
Evaluation	AUC	APR	AUC	APR	AUC	APR
ASAC	0.714 ± 0.018	0.235 ± 0.034	0.781 ± 0.015	0.262 ± 0.037	0.841 ± 0.016	0.414 ± 0.033
DeepSensing	0.707 ± 0.020	0.230 ± 0.036	0.772 ± 0.016	0.255 ± 0.033	0.829 ± 0.017	0.409 ± 0.038
Randomize	0.677 ± 0.021	0.217 ± 0.033	0.729 ± 0.019	0.249 ± 0.032	0.788 ± 0.017	0.269 ± 0.039
SASH	0.725 ± 0.015	0.248 ± 0.032	0.793 ± 0.013	0.278 ± 0.043	0.849 ± 0.014	0.420 ± 0.037

Table 5: *Active Sensing Pathway Example*. Results at different acquisition rates (using GRUs as base predictors).

5 CONCLUSION

Machines will never replace a doctor’s medical judgment, nor an ML researcher’s technical innovation. But as a matter of data-driven *clinical decision support*, Clairvoyance enables rapid prototyping, benchmarking, and validation of complex time-series pipelines—so doctors can spend more time on the real scientific problems, and ML researchers can focus on the real technical questions. Moreover, collaborative research between medical practitioners and ML researchers is increasingly common [48]. To help grease the wheels, we developed and presented Clairvoyance, and illustrated its flexibility and capability in answering important and interesting medical questions in real-world environments.

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